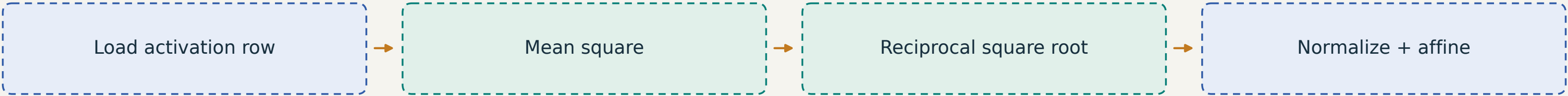


RMSNorm: reduce a row, then reuse its statistics

Standalone normalization · portable Triton historical comparison [N0, N1]

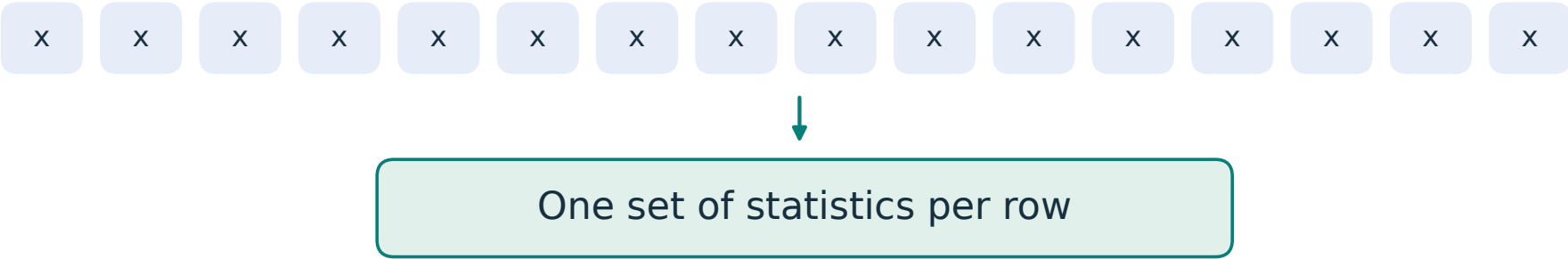
A / EXECUTION AND FUSION



Boxes show the row algorithm. Backward parameter reductions can require additional kernels.

B / ALGORITHM AND DATA REUSE

$$y_i = x_i (\text{mean}(x^2) + \varepsilon)^{-1/2} \gamma_i$$



No centering / mean subtraction.

HISTORICAL MEASUREMENT

Directly measured full operator F+B

L384

0.075 → 0.072 ms

1.04×

L768

0.281 → 0.259 ms

1.08×

Baseline: Pre-change portable Triton
Shape: D128

Measured gains concern the standalone portable path; they do not establish cross-operator CUDA fusion speedups.